

Lecture 15 Worksheet - Transform Dialect for NPU Compiler

A. Handle Annotation

아래 payload IR 에서 transform handle 이 어떤 payload op/value 를 가리키는지 표를 완성하세요.

```
// Payload IR for Lecture 15.
// Goal: schedule MatMul + Bias + ReLU using Transform Dialect.
// This file is intentionally small so that FileCheck-based tests are easy.

#map_matmul_a = affine_map<(m, n, k) -> (m, k)>
#map_matmul_b = affine_map<(m, n, k) -> (k, n)>
#map_matmul_c = affine_map<(m, n, k) -> (m, n)>

module attributes {transform.with_named_sequence} {
  func.func @matmul_bias_relu(
    %A: tensor<128x64xi8>,
    %B: tensor<64x128xi8>,
    %Bias: tensor<128xi32>) -> tensor<128x128xi32> {
    %empty = tensor.empty() : tensor<128x128xi32>
    %zero = arith.constant 0 : i32
    %init = linalg.fill ins(%zero : i32) outs(%empty : tensor<128x128xi32>) -> tensor<128x128xi32>
    %mm = linalg.matmul ins(%A, %B : tensor<128x64xi8>, tensor<64x128xi8>)
      outs(%init : tensor<128x128xi32>) -> tensor<128x128xi32>
    // Bias + ReLU is shown as pseudo linalg.generic for teaching.
    %out = "npu_graph.bias_relu"(%mm, %Bias) : (tensor<128x128xi32>, tensor<128xi32>) -> tensor<128x128xi32>
    return %out : tensor<128x128xi32>
  }
}
```

Transform handle	Payload association	Read/Consume/Produce	Can be used later?
%root			
%matmul			
%tiled_matmul			
%loops			
%epilogue			

B. Tile/Fuse/Vectorize Schedule

아래 빈칸을 채워 16x16x32 native tile 을 사용하는 transform IR schedule 을 작성하세요.

```
transform.named_sequence @_transform_main(%root: !transform.any_op) attributes {transform.readonly} {
  %mm = transform.structured.match ops{["_____"]} in %root
  : (!transform.any_op) -> !transform.any_op
  %tiled, %loops:3 = transform.structured.tile_using_for %mm
    tile_sizes [____, ____, ____]
    : (!transform.any_op) -> (!transform.any_op, !transform.any_op, !transform.any_op, !transform.any_op)
  // Fuse epilogue when legal.
  // Vectorize tiled op.
  transform.structured.vectorize %tiled vector_sizes [____, ____, ____] : !transform.any_op
  transform.yield
}
```

C. NPU Footprint Table

Item	Formula	Value	Fits?
A tile	$M * K * \text{bytes}(A)$		
B tile	$K * N * \text{bytes}(B)$		
ACC tile	$M * N * \text{bytes}(\text{acc})$		
Double buffering	$2 * (A + B) + \text{ACC}$		

D. Failure/Fallback Policy

Case	Failure type	Fallback	Diagnostic message
$K \% 32 \neq 0$			
layout not mk/kn			
dynamic M/N			
SRAM overflow			

E. FileCheck Plan

- CHECK expected loop nest
- CHECK vector.contract shape
- CHECK no unexpected memref.copy
- CHECK negative diagnostic